trained classifier baseline

Orientation for whoever picks this up: what exists, how to run it, what the numbers say, and **which numbers must never be quoted without their caveat**.

- **Status:** committed on `feature/trained-baseline-and-epoch-metrics` (`5ad53d9`), **538 tests passing**, PR not yet opened.
- **The ask:** the supervisor wanted the classic supervised-learning table — precision, recall, F1, support, accuracy *per epoch* — for a model we actually train. Two routes were built, A and B, because GRPO's "training" and a real trained model are different claims.

TL;DR

```
python3 -m pytest tests/ -q                # 538 pass, ~10 s, no Ollama/GPU
python3 -m evaluation.kaggle.grpo_train \
    --episodes evaluation/kaggle/dataset/episodes.jsonl   # Route A table
python3 -m evaluation.train_classifier                     # Route B table
```

Route	What trains	Table	The honest sentence
A GRPO trainer	policy over a 720-action joint grid	`evaluation/kaggle/epoch_metrics.csv` (90 rows: 30 checkpoints × train/holdout)	Curves are **flat** — Phase 5/6 found no reachable threshold gap, so propose-and-verify rejects a no-op. Flat is the correct result, not a bug
B text classifier	TF-IDF + logistic regression (numpy)	`results/classifier/epoch_metrics.csv` + `epoch_metrics_confound_controlled.csv` (60 rows each: 30 epochs × train/val)	Curves rise to 1.000 — but the *cross-corpus* 1.000 is source recognition. Quote the **confound-controlled** one

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1. Route A — per-epoch metrics inside the GRPO trainer

- **"Epoch" = a checkpoint**, every `--metrics-every` steps (default 10, Kaggle-safe), because GRPO scores every step on the full batch — a data pass is not a meaningful unit here. Documented at `evaluation/kaggle/grpo_train.py:624`.
- Columns: `epoch, step, split, n, support_attack, support_benign, tp, fp, fn, tn, precision, recall, f1, accuracy, action` — the `action` column is the policy's joint action at that checkpoint, so each row can be tied back to a concrete configuration.
- Splits are the **address-level holdout**: `HOLDOUT_TARGET_ADDRESSES` (`leaker@shadow-mail.net`, `siphon@nullroute.org`) inlined because the Kaggle bundle cannot import `red_team`; a test pins it to `attack_library.holdout_targets()`.
- The recorder is wired through an **observational `on_step` hook** in all four trainers (`train_torch`, `train_pure`, `train_joint_pure`, `train_joint_torch`). `test_epoch_metrics_hook_does_not_change_the_proposal` proves the chosen action is identical with and without it.
- Last checkpoint per split is copied into `proposed_update.json` under `joint.epoch_metrics` (`{file, checkpoints, every_steps, final{train,holdout}}`).

Measured result (188 episodes, torch/cuda, 30 checkpoints):

split	n	support	first checkpoint (step 10)	last (step 300)
train	140	72 attack / 68 benign	P 0.919 · R 0.944 · F1 0.931 · acc 0.929	identical
holdout	48	48 attack / **0 benign**	1.000 across the board	identical

Do not quote holdout precision. That split is attack-only, so FP is structurally 0 and precision is trivially 1.0 — it is a *recall* split. The false-alarm estimate lives in `evaluation/fpr_report.py` (Wilson,

per cohort).

Also from that run: `risk_threshold` and `window_size` are **unidentifiable** (reward exactly flat — campaigns use a unique `session_id` per case, so the temporal-drift rule never fires), and propose-and-verify **rejected the policy choice as a no-op** (+0.8436 vs +0.8436).

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2. Route B — the trained text-classifier baseline

`baselines/text_classifier.py` (model) + `evaluation/train_classifier.py` (driver). **A comparison baseline, not the defense** — consistent with `FYP/Supervisor_Feedback.md:70`: a classifier judges *wording*, so rewording moves it; the causal gate judges the *action*.

- Corpus: 510 InjecAgent injections vs 68 benign (AgentDojo 60 + our 8 controls). Stratified, seeded (seed 0), disjoint split — 406/86/86; vectorizer fitted on **train only**; epochs 30, `min_df` 2, balanced class weights.
- Reports: per-epoch train+val P/R/F1/macro-F1/accuracy, per-attack-type recall, per-source benign FPR (never pooled), rephrasing probe, optional matplotlib curve.
- The AgentDojo **attack** holdout (`red_team/data/agentdojo_attacks.json`, 253 records, frozen at `46cfbfb`) is read **only** behind `--agentdojo-holdout`, which prints that it is spent for looking and never for tuning. **Not spent.**

Cross-corpus test set (n = 86: 76 attacks / 10 benign) — all 1.000: detection 76/76 [0.952, 1.000]; false alarms 0/10 [0.000, 0.278]; recall by type 1.000 for Data Security Harm (27), Financial Harm (23), Physical Harm (26); benign FPR `agentdojo_benign` 0/9 [0.00, 0.30], `ours_benign` 0/1 [0.00, 0.79].

Confound-controlled cohort (the one to quote) — InjecAgent vs its own **de-injected twins**: each original with its `attacker_instruction` span deleted (510 pairs, 0 dropped; split by *pair* so an original never sits beside its own copy — 358/76/76). Twin test n = 152 (76/76), all 1.000. A twin is its original minus the imperative, so the only features left that separate the classes **are the injection's own words** — this 1.000 means detection, not source.

Rephrasing probe (same attacks, rewritten; `kept` = share of clean tokens surviving):

row	cross-corpus recall (kept)	twin recall (kept)
clean	1.000 (1.000)	1.000 (1.000)
paraphrase — hand-built synonym rewrite	1.000 (0.869) → **+0.000**	0.921 (0.886) → **−0.079**
datamarking	1.000 (1.000)	1.000 (1.000)
delimiting	1.000 (1.000)	1.000 (1.000)
encoding	1.000 (0.000)	1.000 (0.000)
empty-input control	1.000 (0.000)	1.000 (0.000)

Read it as: the first table never looks at the words (reword 13% of them and nothing moves → source recognition); the twin model does, and rewording costs it 0.079; spotlighting's character transforms cost **0.000** because a character-class tokenizer re-derives every token — they change the surface a *reader* sees, not the words this model sees; and `encoding` scoring like an empty input is the **intercept** (`b = +0.096 > 0`, so content-free input defaults to "attack"), not a detection.

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3. Findings that changed the code — read before quoting anything

1. **Three test failures were my hand-arithmetic, not the code.** `support_attack = tp+fn`, `support_benign = tn+fp`; the classifier test's vectors counted (2,1,2,5); `macro_f1 = 0.5` on an all-benign-correct `[0,0]/[0,0]` (which is exactly why macro-F1 must not be quoted when a class is absent). All corrected with exact fractions in the assertions.

2. **The twin cohort had its own label leak.** Deleting the instruction left `''` behind, and the tokenizer kept apostrophes — so `''` was the strongest feature in the model (**−1.07**): it had learned *how the twins were built*. Fixed by a punctuation-blind tokenizer (`[a-z0-9]+`), with the reasoning in `baselines/text_classifier.py` and recorded in the manifest. Chosen from train/val inspection; **the frozen attack holdout was never read to make the choice.**
3. **The probe's prose contradicted its own table** — a sentence claiming rewording moves the model printed directly above a table showing `0.000`, twice. Fixed by `print_probe_verdict()`, which recomputes every printed number from the rows above it, pinned by `test_probe_verdict_prints_the_numbers_it_was_given`.
4. **`encoding = 1.000` was the intercept, not detection** (zero features left, positive bias). Fixed by the empty-input control row.

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4. How to run / regenerate

Command	Output	Cost
`python3 -m pytest tests/ -q`	538 pass	~10 s, no LLM/network/GPU
`python3 -m evaluation.train_classifier`	`results/classifier/*` (5 files)	<1 min CPU
`python3 -m evaluation.train_classifier --epochs 40 --seed 1`	same, varied	idem
`python3 -m evaluation.train_classifier --agentdojo-holdout`	spends the holdout — **requires a decision, not a routine flag**	idem
`python3 -m evaluation.kaggle.grpo_train --episodes evaluation/kaggle/dataset/episodes.jsonl`	`evaluation/kaggle/epoch_metrics.csv` + `proposed_update.json`	~1 min (torch/cuda if present)
`python3 -m baselines.text_classifier`	tiny self-test on synthetic data	instant

`evaluation/kaggle/epoch_metrics.csv` is **gitignored** (regenerated); the `results/classifier/` artifacts are **committed** with their manifest.

> The committed `manifest.json` stamps `commit 44adf79, dirty: true` because it records HEAD *at run time* — the work was committed afterwards as `5ad53d9`. Re-running the driver after the merge will stamp the real commit.

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5. Tests

`python3 -m pytest tests/ -q` → **538 passed** (478 pre-existing + 33 in `tests/test_grpo_kaggle.py` + 27 in `tests/test_text_classifier.py`).

New coverage worth knowing about:

- confusion matrix agrees with the reward's own catch predicate; the two `classification_metrics` copies (`grpo_env` vs `text_classifier`) agree on the same matrix (they must — two copies of one formula is how drift starts);
- the epoch hook is **observational** (same chosen action with and without);
- address split pinned to `attack_library`; the holdout file is not opened on the default code path (a `builtins.open` spy);
- the twin corpus really removes the injection (a twin still carrying it is a label error) and its pair-split keeps originals beside their twins;
- tokenizer is punctuation-blind; `paraphrase` substitutes rather than deletes; the probe reports all six rows with token-retention;
- the printed verdict is derived from the rows it prints.

6. File map

File	Role
`evaluation/kaggle/grpo_train.py`	Route A: `HOLDOUT_TARGET_ADDRESSES`, `episode_split`, `epoch_metrics`, `_epoch_recorder`, `write_epoch_metrics`, `print_epoch_summary`, `on_step` hooks, `--metrics-every`
`evaluation/kaggle/grpo_env.py`	`confusion_counts` / `classification_metrics` (stdlib-only)
`baselines/text_classifier.py`	`TfidfVectorizer`, `LogisticClassifier` (balanced weights, `eval_set` → per-epoch rows), `classification_metrics`, punctuation-blind `tokenize`
`evaluation/train_classifier.py`	Driver: loaders, `load_injecagent_twin_corpus`, `split_pairs`, `twin_docs`, `REPHRASE_MAP`/`paraphrase`, `rephrasing_probe`, `print_probe_verdict`, report + manifest writers
`tests/test_grpo_kaggle.py`	33 tests (10 new)
`tests/test_text_classifier.py`	27 tests, new file
`results/classifier/`	Committed artifacts: 2 metric CSVs, report JSON, manifest JSON, curve PNG
Docs updated	`README.md`, `Architecture.md`, `evaluation/README.md`, `evaluation/kaggle/README.md`, `results/README.md`, `tests/README.md`, `.gitignore`

7. Rules compliance (docs/Rules.md)

- **§6** — every rate carries `n` + Wilson interval; recall reported per attack type; the two benign cohorts never pooled; cross-corpus vs twin results reported as separate tables, never blended; corpus–label confound stated in the report itself, not left for a reader to find.
- **Holdout** — `agentdojo_attacks.json` unopened by default, guarded by a test; scoring it is opt-in and announces that it is spent for looking.
- **§8** — branch + PR, never straight to `main`.
- **§159** — never describe the GRPO policy as "trained"; "epoch" is defined as a checkpoint and labelled as such.

8. Git state & next actions

```
branch: feature/trained-baseline-and-epoch-metrics   commit 5ad53d9
main:   44adf79 (Add frontend design…)
```

Committed: 15 files (code, tests, docs, `results/classifier/`). Deliberately **left uncommitted in the working tree — yours, not mine**: `FYP/*.pptx`, `FYP/ppt skills/`, the `.drawio` move to `docs/`.

Next:

```
git push -u origin feature/trained-baseline-and-epoch-metrics
gh pr create --title "Add per-epoch metrics to the GRPO trainer and a trained classifier baseline" \
  --body "Supervisor's supervised-learning table: per-epoch P/R/F1/support/accuracy for both the
GRPO trainer and a trained TF-IDF + logistic-regression baseline. 538 tests pass."
```

9. Known limitations & open decisions

1. **Twin negatives are synthetic** (instruction span deleted). Deletion is span-level and verbatim (0 of 510 dropped), but a future reader should know the twins are constructed, not naturally benign documents.
2. **The paraphrase map is hand-built and partial** (~13% of tokens touched) — read `−0.079` as a **lower bound** on a real rephraser. An LLM paraphrase probe would be the stronger test and needs an offline/deterministic plan to stay inside the test rules.
3. **Encoding/empty-input rows are the intercept.** If `b` ever flips negative, those rows flip too — the control row is what keeps them honest.
4. **GRPO's holdout split is attack-only** (see §1) — precision there is decorative; do not report it.
5. **Spending the attack holdout** (`--agentdojo-holdout`) is a one-way door. Decide with the supervisor, run once, record the number with its `n` + Wilson, and never tune after seeing it.
6. **`risk_threshold` / `window_size` are unidentifiable on this data** (flat reward) — if anyone wants those learned, the campaign needs a reused `session_id` so the temporal-drift rule can fire; that is a data change, not a trainer change.